

OpenWrt 25.12.5 Build Guide for Banana Pi BPI-R4-8P

A Practical Step-by-Step Guide

- Status: Fully validated
- Wireless: Banana Pi BPI-R4-NIC-BE14 (MediaTek MT7996)
- Linux Kernel: 6.12
- Profile: Banana Pi BPI-R4 PoE
- Target: mediatek / filogic
- Board: Banana Pi BPI-R4-8P
- OpenWrt: 25.12.5

Document Information

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Platform: Banana Pi BPI-R4-8P

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Tested Configuration

Introduction

This guide describes the complete process of building a customized OpenWrt 25.12.5 firmware image for the Banana Pi BPI-R4-8P platform. The document is based on a verified working build and reflects the actual configuration used during development.

Unlike a generic build tutorial, every step in this guide has been validated on real hardware, including source preparation, build configuration, application of the MT7996 patch, firmware compilation, initial boot, BL2 installation, and eMMC expansion.

Target Hardware

- Board: Banana Pi BPI-R4-8P
- SoC: MediaTek MT7988
- Wireless module: Banana Pi BPI-R4-NIC-BE14 (MediaTek MT7996)

- Memory: 8 GB LPDDR4
- Storage: 8 GB eMMC

Software Environment

- Host operating system: Ubuntu 24.04 LTS
- OpenWrt release: 25.12.5
- Linux kernel: 6.12
- Target: mediatek / filogic
- Target profile: Banana Pi BPI-R4 PoE (DEVICE_bananapi_bpi-r4-poe)

Scope

The guide assumes a clean OpenWrt source tree and focuses on producing a reproducible firmware image. Only one additional source patch is applied: the MT7996 transmit power patch imported from ImmortalWrt. All remaining configuration changes are performed using the standard OpenWrt build system.

Part 02. Preparing the Build Environment

2.1 Host Operating System

Ubuntu LTS is recommended as the build host. All commands in this guide are intended to be executed by a regular user with sudo privileges.

2.2 Installing Build Dependencies

Update the package index and install the required development tools:

```
sudo apt update
```

```
sudo apt install -y build-essential clang flex bison g++ gawk gcc-multilib  
gettext git libncurses-dev libssl-dev python3 python3-distutils rsync unzip  
zlib1g-dev file wget curl
```

2.3 Verifying the Toolchain

Verify that the required tools are available:

```
git --version
```

```
gcc --version
```

```
python3 --version
```

```
make --version
```

Verification

Verify the following before proceeding:

- the operating system is fully updated;
- all required packages have been installed successfully;
- Git, GCC, Python and Make report valid version information.

Part 03. Obtaining the OpenWrt Source Tree

3.1 Cloning the Official Repository

Create a working directory, clone the official OpenWrt repository, and switch to the verified release tag used throughout this guide.

```
mkdir -p ~/openwrt
```

```
cd ~/openwrt
```

```
git clone https://github.com/openwrt/openwrt.git
```

```
cd openwrt
```

```
git checkout v25.12.5
```

3.2 Updating and Installing Feeds

Update all package feeds and install their contents before configuring the build.

```
./scripts/feeds update -a
```

```
./scripts/feeds install -a
```

3.3 Downloading the Official Build Configuration

Download the official build configuration for the OpenWrt 25.12.5 mediatek/filogic target:

```
wget
```

```
https://downloads.openwrt.org/releases/25.12.5/targets/mediatek/filogic/config.buildinfo -O official.config
```

Verify that the file has been downloaded successfully:

```
ls -lh official.config
```

3.4 Preparing the Working Configuration

Use the official configuration as the baseline for the customized build:

```
cp official.config .config
```

```
make defconfig
```

Verification

Verify the following before proceeding:

- the repository is checked out at tag v25.12.5;
- all feeds have been updated and installed successfully;
- the file official.config is present in the project directory;
- the .config file has been generated successfully using make defconfig.

Part 04. Configuring the Build

4.1 Launching menuconfig

Change to the OpenWrt source directory and launch the configuration utility:

```
cd ~/openwrt/openwrt
```

```
make menuconfig
```

Configure the following target options:

```
Target System ---> MediaTek Ralink ARM
```

```
Subtarget ---> Filogic
```

```
Target Profile ---> Banana Pi BPI-R4 PoE
```

Save the configuration and verify that the correct target profile has been selected:

```
grep -E "CONFIG_TARGET_|bananapi" .config | grep -v "^#"
```

The output should include:

CONFIG_TARGET_mediatek=y

CONFIG_TARGET_mediatek_filologic=y

CONFIG_TARGET_mediatek_filologic_DEVICE_bananapi_bpi-r4-poe=y

CONFIG_TARGET_PROFILE="DEVICE_bananapi_bpi-r4-poe"

4.2 Additional Packages Enabled

Compared to the official configuration (official.config), the following packages were additionally enabled for this build:

- kmod-tun
- kmod-inet-diag
- kmod-nft-tproxy
- block-mount
- kmod-fs-f2fs
- e2fsprogs
- f2fs-tools
- mkf2fs
- f2fsck
- mmc-utils
- openssh-sftp-server
- curl
- mc
- nano-full
- htop
- iperf3
- pciutils
- usbutils
- ethtool-full
- cfdisk
- tcpdump
- lsof
- strace
- kmod-wireguard
- wireguard-tools

Verification

Verify the following before proceeding:

- the Banana Pi BPI-R4 PoE target profile is selected;
- the updated .config file has been saved successfully;
- all additional packages listed above are enabled.

Part 05. Applying the MT7996 Transmit Power Patch

5.1 Overview

The Banana Pi BPI-R4-8P used in this guide is equipped with the Banana Pi BPI-R4-NIC-BE14 wireless module based on the MediaTek MT7996 chipset. This build uses a single additional source modification: the MT7996 transmit power patch imported from the ImmortalWrt project.

5.2 Downloading the Patch

Patch source:

```
https://github.com/immortalwrt/immortalwrt/blob/master/package/kernel/mt76/patches/100-wifi-mt76-mt7996-Use-tx\_power-from-default-fw-if-EEP.patch
```

Download the patch directly into the OpenWrt source tree:

```
wget  
https://raw.githubusercontent.com/immortalwrt/immortalwrt/master/package/kernel/mt76/patches/100-wifi-mt76-mt7996-Use-tx\_power-from-default-fw-if-EEP.patch -P package/kernel/mt76/patches/
```

Verify that the patch file is present:

```
ls package/kernel/mt76/patches/
```

5.3 Using a Local Copy

If the patch has already been downloaded directly into package/kernel/mt76/patches/, this step may be skipped.

Otherwise, copy the local patch file into the patch directory:

```
cp ~/patches/100-wifi-mt76-mt7996-Use-tx_power-from-default-fw-if-EEP.patch package/kernel/mt76/patches/
```

Verification

Verify the following before proceeding:

- the file 100-wifi-mt76-mt7996-Use-tx_power-from-default-fw-if-EEP.patch is present in package/kernel/mt76/patches/;
- no other source code modifications have been applied.

Part 06. Building OpenWrt

6.1 Preparing the Build

Before starting the compilation, synchronize the working configuration with all selected packages and dependencies.

```
make defconfig
```

6.2 Compiling the Firmware

Run the standard parallel build:

```
make -j$(nproc)
```

If the build fails, repeat the compilation in verbose single-threaded mode to simplify troubleshooting:

```
make -j1 V=s
```

6.3 Build Time

Compilation time depends on the performance of the build host. Avoid interrupting the process once compilation has started.

6.4 Verifying Build Results

When the build completes successfully, verify that the firmware images have been generated in the target output directory.

```
ls -lh bin/targets/mediatek/filogic/
```

The output should include the firmware images for the Banana Pi BPI-R4 PoE target, including the initramfs image and the storage-specific installation images.

Verification

- the build completed without errors;
- firmware images are present in `bin/targets/mediatek/filogic/`;
- all generated image files have a valid, non-zero size.

Part 07. Flashing, Initial Boot and System Verification

7.1 Locating the Firmware Images

After a successful build, the generated firmware images are located in:

```
bin/targets/mediatek/filogic/
```

Select the image that matches the intended installation method (initramfs, NAND or eMMC).

7.2 First Boot

Flash the appropriate firmware image and boot the system. After the first boot, verify the installed OpenWrt release:

```
cat /etc/openwrt_release
```

7.3 Verifying the Hardware

Verify that all network interfaces have been detected:

```
ip link
```

Verify that the wireless interfaces are present:

```
iw dev
```

Verify the capabilities of the Banana Pi BPI-R4-NIC-BE14 wireless module:

```
iw list
```

The output should confirm that the MT7996-based wireless module has been initialized correctly and that the supported radio bands and Wi-Fi 7 (EHT) capabilities are available.

Verify the available system memory:

```
free -h
```

7.4 Verifying System Services

Review the system status and recent log messages to confirm normal operation:

ubus list

logread | tail -50

dmesg | tail -50

Verification

- the system boots successfully;
- all expected network interfaces are available;
- the Banana Pi BPI-R4-NIC-BE14 wireless module is detected correctly;
- 8 GB of system memory is available;
- no critical errors are reported in the system log.

Part 08. Installing BL2, Enabling 8 GB RAM and Expanding eMMC

8.1 Preparing the BL2 Bootloader

To enable full support for 8 GB of system memory, a modified BL2 bootloader must be installed before booting OpenWrt from eMMC.

Download the following file:

bpi-r4_emmc_8GB_bl2.img

Official source:

<https://github.com/frank-w/u-boot/releases>

For consistency throughout this guide, rename the downloaded file:

mv bpi-r4_emmc_8GB_bl2.img bl2_emmc_8g.img

8.2 Writing BL2 to eMMC

Use the NAND-based recovery firmware prepared earlier. It must include the mmc-utils and openssh-sftp-server packages required to transfer the BL2 image and write it to the eMMC boot partition.

Copy bl2_emmc_8g.img to /tmp on the target device, then execute:

```
echo 0 > /sys/block/mmcblk0boot0/force_ro
```

```
dd if=/tmp/bl2_emmc_8g.img of=/dev/mmcblk0boot0
```

```
mmc bootpart enable 1 1 /dev/mmcblk0
```

After writing BL2, boot the system from the eMMC installation.

8.3 Expanding the eMMC Partition

Boot the NAND recovery firmware and start the partition editor:

```
cfdisk /dev/mmcblk0
```

In cfdisk, perform the following steps:

1. Select the /dev/mmcblk0p5 partition.
2. Choose Resize and expand the partition to the maximum available size.
3. Select Write.
4. Confirm the operation by typing "yes".
5. Reboot into the OpenWrt installation on eMMC.

8.4 Final Verification

Verify that the upgrade completed successfully:

```
free -h
```

```
lsblk
```

Verification

- the system reports 8 GB of installed memory;
- the root partition has been expanded successfully;
- the device boots correctly from eMMC.

Appendix A. Build Profile

The following table summarizes the validated hardware and software configuration used throughout this guide.

Parameter	Value
Board	Banana Pi BPI-R4-8P
Target Profile	Banana Pi BPI-R4 PoE (DEVICE_bananapi_bpi-

	r4-poe)
SoC	MediaTek MT7988
Memory	8 GB LPDDR4
Storage	8 GB eMMC
Wireless Module	Banana Pi BPI-R4-NIC-BE14 (MediaTek MT7996)
OpenWrt Release	25.12.5
Linux Kernel	6.12
Additional Source Patch	100-wifi-mt76-mt7996-Use-tx_power-from-default-fw-if-EEP.patch
Additional Source Modifications	None

Appendix B. Post-build Checklist

Before deploying the firmware, verify the following:

- The firmware was built successfully.
- The MT7996 patch has been applied.
- The system boots correctly.
- 8 GB of RAM is detected.
- The eMMC partition has been expanded.
- The Banana Pi BPI-R4-NIC-BE14 wireless module is detected by 'iw list'.
- No critical errors are present in the system log.

Appendix C. References

The following resources were used during the preparation and validation of this guide:

- OpenWrt Project — <https://github.com/openwrt/openwrt>
- OpenWrt 25.12.5 Release — <https://downloads.openwrt.org/releases/25.12.5/>
- ImmortalWrt Project — <https://github.com/immortalwrt/immortalwrt>
- Frank-W U-Boot Releases (BL2) — <https://github.com/frank-w/u-boot/releases>

Document Revision

This document reflects a validated OpenWrt 25.12.5 build for the Banana Pi BPI-R4-8P platform. Future OpenWrt releases may introduce changes to the build system, package selection, or installation process. The procedures described in this guide are fully reproducible for the software versions documented herein.

Acknowledgements

This guide was developed, validated, and documented using real hardware and a fully reproducible build environment.

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